

Element C student work

Thursday, September 25, 2025 12:55 PM

Design Criteria

Note: AI was used to generate ideas for criteria and justifications

- **Safety First**

The device must not harm the turtle in any way—physically, emotionally, or behaviorally. This includes avoiding sharp edges, toxic materials, small parts that could be ingested, or mechanisms that could trap or stress the turtle. If the turtle is harmed in any way by the device after 6 months of sustained use, the device is to be considered a failure.

Note: This is the **highest priority criteria**, and the following are listed in decreasing priority.

Justification: Turtles are sensitive creatures, and any physical harm—like cuts, ingestion of small parts, or entrapment—can lead to serious health issues. Ensuring safety protects their well-being and prevents stress or injury.

- **Species-Appropriate Engagement**

It should cater to the natural behaviors of turtles, such as exploring, basking, or burrowing. The enrichment should encourage activity without causing confusion or frustration. The turtles should be observable interacting with the device in the intended manner.

Justification: Enrichment should reflect natural turtle behaviors such as basking or exploring. This keeps them mentally stimulated and supports their physical health by encouraging movement and curiosity.

- **Durability and Cleanability**

The device must be sturdy enough to withstand repeated use and easy to clean to prevent the buildup of bacteria or mold, which could pose health risks. The device must be made from materials that will suffer little to no degradation over a period of 6 months of continuous use. Materials rated for maritime use, like heavy duty rope, certain plastics, etc. are preferable.

Justification: Turtles often interact with objects using their mouths or claws, so the device must withstand wear and tear. Easy cleaning is essential to prevent bacterial growth, which could lead to infections or illness.

- **Environmental Compatibility**

It should be suitable for the turtle's habitat, and not disrupt temperature, humidity, or water quality. If waterborne, the device must be able to sit in water for 6 months without releasing a measurable amount of any contaminant. Additionally, the device must be able to fit into a 2' by 4' by 1' space.

Justification: The device must not interfere with the turtle's habitat conditions—like water quality, temperature, or humidity. Compatibility ensures the turtle's environment remains stable and healthy.

- **Variety and Stimulation**

The device should offer sensory or cognitive stimulation to prevent boredom and promote mental well-

being. This could include changing textures, colors, or interactive elements. For the device to be considered successful, the turtles must demonstrate increases in active exploration, play behavior, and engagement in enrichment activities, which are indicative of wellness.

Justification: Changing textures, shapes, or interactive features can prevent boredom and promote cognitive development. Mental stimulation is key to a turtle's overall well-being, especially in captivity.

Effect on the Stakeholders

A device which improves the quality of life of the turtles, will also improve the quality of life of the primary stakeholders. Boredom, or lack of enrichment, is shown to lead to an increase in stress and depression for turtles. By decreasing the likelihood of the turtles suffering from stress or depression, this device will improve the quality of the turtles for the purposes of research or exhibition. Additionally, the device has the potential to reduce the need for human enrichment, such as feeding the turtles crickets, which decreases the time commitment to maintaining the turtles.

Note from Mr Martz:

You have shown some very good work here. C3 was your weak point – just list them as high priority!

Note from Mr Martz 11/6 - good additions to your work!